

Project Documentation: Land Type Classification in Greater Cairo using Machine Learning

1. Project Planning & Management

Project Proposal

Overview: This project focuses on classifying land use and land cover (LULC) in the Greater Cairo region using multispectral satellite imagery from the Sentinel-2 mission. By leveraging Deep Neural Networks (DNNs), the system aims to accurately identify various land types, including urban areas, agricultural land, water bodies, desert, roads, and vegetation.

Objectives:

- Develop a robust Deep Learning model for land type classification.
- Apply the model specifically to the Greater Cairo region to aid urban planning and environmental monitoring.
- Provide a scalable solution for processing multispectral satellite data. **Scope:** The project covers data collection (Sentinel-2), preprocessing, exploratory data analysis (EDA), model development (CNN/DNN), evaluation, and deployment as a functional API or web service.

Task Assignment & Roles

- **Data Scientist / ML Engineer:** Responsible for data preprocessing, model architecture design, training, and evaluation.
- **GIS Specialist:** Responsible for handling satellite imagery, coordinate systems, and manual labeling using QGIS if necessary.
- **Full-Stack Developer:** Responsible for deploying the model as a web service and creating the user interface.

Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation Strategy
Cloud Cover in Images	High	Use multi-temporal images or atmospheric correction techniques.
Class Imbalance	Medium	Apply data augmentation and weighted loss functions during training.
High Computational Cost	Medium	Use transfer learning with pre-trained models (e.g., EuroSat) to reduce training time.
Data Labeling Errors	High	Cross-validate labels using QGIS and ground-truth data.

KPIs (Key Performance Indicators)

- Classification Accuracy:** Target > 85% across all land types.
- F1-Score:** High balance between precision and recall for critical classes like "Urban" and "Agriculture".
- Inference Time:** Model should process a standard satellite tile in under 2 seconds.
- User Adoption:** Successful integration into a web-based visualization tool.

2. Literature Review

The project builds upon existing research in remote sensing and deep learning. Key references include the **EuroSat dataset** for land use classification and various architectures like **ResNet** and **U-Net** which have shown superior performance in pixel-wise and patch-based classification tasks. The use of **NDVI (Normalized Difference Vegetation Index)** is a standard practice for distinguishing vegetation from other land types.

3. Requirements Gathering

Stakeholder Analysis

- Urban Planners:** Need accurate maps to monitor city expansion in Greater Cairo.
- Environmental Agencies:** Require data on desertification and loss of agricultural land.

- **Researchers:** Interested in the application of DNNs on multispectral data.

User Stories

- *As an urban planner,* I want to upload a satellite image of a specific district in Cairo so that I can see the percentage of built-up areas versus green space.
- *As an environmentalist,* I want to track changes in the Nile's surrounding vegetation over time.

Functional Requirements

- The system must support Sentinel-2 multispectral image formats (GeoTIFF).
- The model must classify at least 6 distinct land types.
- The system must provide a visualization of the classification results (heatmaps/labeled maps).

Non-functional Requirements

- **Performance:** High-speed processing of large satellite tiles.
 - **Scalability:** Ability to handle multiple concurrent classification requests.
 - **Usability:** Simple web interface for non-technical users to upload and view results.
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4. System Analysis & Design

Problem Statement & Objectives

Greater Cairo is experiencing rapid urban growth, often at the expense of agricultural land and natural resources. Manual classification is slow and prone to error. This project aims to automate this process using AI to provide timely and accurate land-use data.

Software Architecture

The system follows a **Client-Server Architecture**:

- **Frontend:** React/TailwindCSS for the user interface.
- **Backend:** FastAPI/Flask to handle image uploads and serve model predictions.
- **Model Engine:** TensorFlow/PyTorch running the trained CNN model.

Database Design & Data Modeling

- **Metadata Storage:** Storing image Metadata, classification results, and user history.

- **Schema:** Tables for Users, Images, Classification Results, and Land Types.

Data Flow & System Behavior

- 1 **User** uploads a Sentinel-2 image.
- 2 **Backend** preprocesses the image (resizing, normalization, NDVI calculation).
- 3 **ML Model** performs inference.
- 4 **The system** generates a classified map and statistics.
- 5 **User** views and downloads the report.

UI/UX Design

- **Dashboard:** A map-centric view showing the Greater Cairo region.
 - **Upload Portal:** Drag-and-drop zone for satellite data.
 - **Results Panel:** Side-by-side comparison of original vs. classified images.
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5. Implementation (Source Code & Execution)

Source Code Standards

- **Modular Design:** Separate modules for data loader, model trainer, preprocessor, and Api handler.
- **Documentation:** All functions and classes are documented using Google-style docstrings.
- **Version Control:** Hosted on GitHub with a clear branching strategy (main, develop, feature-branches).

Deployment & Execution

- **Environment:** Python 3.11 with dependencies managed via requirements.txt.
 - **Execution:** The project can be run locally using uvicorn main: app or via Docker containers.
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6. Testing & Quality Assurance

- **Unit Testing:** Testing individual preprocessing functions and model layers.
- **Integration Testing:** Ensuring the API correctly communicates with the ML model.

- **Validation:** Using a hold-out test set from the Greater Cairo region to verify real-world performance.
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7. Final Presentation & Reports

- **User Manual:** Detailed guide on how to use the web interface and interpret classification maps.
- **Technical Report:** Deep dive into model hyperparameters, loss curves, and confusion matrices.
- **Presentation:** A slide deck summarizing the project's impact on urban monitoring in Egypt.